

Alex Siegel

1200 E. California Blvd, MC 147-75, Pasadena, CA 91125
(818) 439-7884 | asiegel@caltech.edu | <http://alexrsiegel.github.io>

SUMMARY

I'm a research scientist with over 10 years of experience applying biophysical and biochemical techniques to study the mechanisms of proteins. My work offers a detailed mechanistic understanding of how protein chaperones prevent the aggregation of their misfolding clients and provides a platform to build custom chaperones to combat aggregation-related diseases. I'm looking for a position where I can apply my expertise to study the molecular mechanisms of proteins associated with diseases and contribute to the development of novel therapeutics.

EDUCATION

University of California, Berkeley

Ph.D. in Biophysics with specialization in Chemical Biology
Graduate advisor – Prof. David E. Wemmer, Chemistry
Thesis: Mechanisms of σ^{54} bacterial transcription activation

Berkeley, CA
June 2016

California Institute of Technology

B.S. in Biology and Chemistry, double major, graduated with Honors

Pasadena, CA
June 2008

RESEARCH EXPERIENCE

*Indicates equal contribution first author publications.

Postdoctoral Scholar in Chemistry at Caltech

Research with Prof. Shu-ou Shan

Pasadena, CA
2017-present

- Mapped interactions between a critical plant chaperone, cpSRP43, and membrane protein LHCPs that assemble into photosynthesis machinery by crosslinking-coupled mass spectrometry. ([McAvoy, Siegel, et al, JBC, 2018](#)).
- Identified a new class of cpSRP43 clients involved in chlorophyll synthesis ([Wang et al, PNAS, 2018](#)) and characterized cpSRP43's novel thermoprotection activity towards them ([Ji*, Siegel* et al, Nature Plants, 2021](#)).
- Discovered a unique activation mechanism triggered by an order-to-disorder transition ([Siegel*, McAvoy* et al, JMB, 2020](#)) and showed how cpSRP43 can switch from membrane protein LHCPs to globular protein clients ([Siegel et al, In Revision at Science Advances, 2025](#)) or to non-native amyloid-forming clients ([Prov. Patent ID: CIT-9049-P](#)).
- Solved the cryo-EM structure of human mitochondrial ClpB homolog, Skd3, revealing its two roles as a disaggregase chaperone and as a chaperonin-like folding cage ([Gupta*, Lentzsch*, Siegel*, et al, Science Advances, 2023](#)).

Graduate Researcher in Biophysics at UC Berkeley

Research with Prof. David Wemmer

Berkeley, CA
2009-2016

- Revealed how NtrC transcriptional activators initiate transcription by threading the N-terminus of an RNA Polymerase subunit through their central pore to apply the force needed to melt DNA ([Siegel et al, JMB, 2016](#)).
- Used molecular tweezers to understand the force-dependent activation of RNA polymerase ([Siegel, dissertation](#)).

PUBLICATIONS

*Indicates equal contribution first author publications.

- **Siegel A**, Kroon G, Zhao CQ, Wang P, Wright PE, and Shan SO. Switchable client specificity in a dual functional chaperone coordinates light harvesting complex biogenesis. In Revision at [Science Advances](#) (2025).
- **Siegel A***, Singh J*, Qin PZ & Shan SO EPR Studies of Chaperone Interactions and Dynamics. Biophysics of Molecular Chaperones (eds. Hiller, S., Liu, M. & He, L.) 242–277. [Royal Society of Chemistry](#) (2023).
- Gupta, A*, Lentzsch AM*, **Siegel A***, Yu Z, Chio US, Cheng Y, & Shan SO. Dodecamer assembly of a metazoan AAA+ chaperone couples substrate extraction to refolding. [Science Advances](#). 9(19), eadf5336 (2023).
- Ji S*, **Siegel A***, Shan, S., Grimm, B. & Wang, P. Chloroplast SRP43 autonomously protects chlorophyll biosynthesis proteins against heat shock. [Nature Plants](#). 7, 1420-1432 (2021).
- **Siegel A***, McAvoy C*, Lam V, Liang FC, Kroon G, Miaou E, Griffin P, Wright P, Shan SO. A Disorder-to-Order Transition Activates an ATP-Independent Membrane Protein Chaperone. [Journal of Molecular Biology](#). 432, 166708 (2020).

- McAvoy C, **Siegel A**, Piszkiwicz S, Miaou E, Yu M, Nguyen T, Moradian A, Sweredoski MJ, Hess S, Shan SO. Two Distinct Sites of client protein interaction with the chaperone cpSRP43. *Journal of Biological Chemistry*. RA118, (2018).
- Wang P, Liang FC, Wittmann D, **Siegel A**, Shan SO, Grimm B. Chloroplast SRP43 acts as a chaperone for glutamyl-tRNA reductase, the rate-limiting enzyme in tetrapyrrole biosynthesis. *Proceedings of the National Academy of Sciences*. 115(15), E3588-96 (2018).
- **Siegel AR**, Wemmer DE. Role of the $\sigma 54$ Activator Interacting Domain in Bacterial Transcription Initiation. *Journal of Molecular Biology*. 428(23), 4669-85 (2016).
- **Siegel AR**, Mechanisms of $\sigma 54$ bacterial transcription activation. Doctoral Dissertation University of California, Berkeley. Publication No. 10192026. ProQuest Dissertations Publishing (2016).

PATENTS

Patent ID: CIT-9049-P.

Filed in 2023

Provisional patent application for use of chaperones against neurodegenerative diseases.

SKILLS

*Experimental or computational skills with proficiency (instruments/software/methods used)

- Independent researcher that can initiate original projects in new areas from gaps in primary literature.
- Scientific communicator to diverse audiences both at conference presentations and through drafting published papers.
- Directly trained over 20 young researchers, including roles as a project leader for group research.
- Lengthy experience with a wide range of techniques in biophysics, biochemistry, and molecular biology (see below).

NMR Spectroscopy. Studying protein structure and dynamics by NMR Spectroscopy.

- Isotope labeling (M9 media growth)
- Spectra collection (Bruker Topspin)
- Writing pulse programs (Bruker Topspin)
- Data analysis (NMRpipe, NMRview, CARA, MestreNova)
- Spectrometer maintenance (Bruker 600-900 MHz, nitrogen & helium fills, basic maintenance).
- Site-specific labeling (MTSL & MMTS, ^{19}F -Phe, BTFA)
- Multiple dimensional collection and analysis (Diffusion ordered spectroscopy, triple resonance assignments, structure determination by NOE)
- Paramagnetic relaxation enhancement
- Non-uniform sampling

Biophysics Techniques. Biophysical techniques to characterize protein structure, dynamics, and interactions.

- X-Ray Crystallography
- Mass Spectrometry (Orbitrap Elite, Q-Exactive)
- Electron Paramagnetic Resonance (Bruker CW-EPR)
- Cryo Electron Microscopy (Titan Krios)
- Small Angle X-ray Scattering
- Fluorescence Anisotropy and FRET (Horiba Fluorolog)
- Single molecule FRET (ALEX, Hamamatsu Photonics picosecond streak camera)
- Fluorescence lifetime measurements (Horiba TCSPC)
- Mass Photometry (Refeyn)
- Circular Dichroism (Aviv model 410)
- Molecular Tweezers

Molecular Biology Techniques. Cloning, purifying, and *in vitro* assays to study proteins.

- Cloning (site-directed mutagenesis, Gibson Assembly)
- Protein expression (E. coli)
- PCR/qPCR
- Protein purification (FPLC/Chromatography)
- Immunoassays (western blotting and quantification, immunoprecipitation)
- Amber suppression (Schultz method)
- In vitro translation (PURE and self-prepared systems)

Biochemistry Techniques. Labeling and other biochemical assays to monitor protein activity.

- Light scattering (Beckman DU series)
- Luciferase folding assays (SpectraMax iD5 plate reader)
- ATPase assays
- Chemical crosslinking (BMOE, BMH, DSSO, DHSO)
- Fluorescence labeling (AlexaFluor/BODIPY maleimides)
- EPR and NMR labeling (MTSL)
- Chromatography (Agilent 1100 series HPLC).

Software. Experience with various protein structure packages, bioinformatics tools, and coding languages.

- Structure visualization (ChimeraX, Pymol)
- Cryo-EM analysis (CryoSPARC, Relion)
- Protein structure determination (PHENIX, Coot)
- Mass spectrometry (ProteinProspector, MeroX)
- Structure computation (Gromacs, AlphaFold 2)
- Bioinformatics (BioPython)
- Data analysis (R, Matlab, Graphpad Prism)
- Coding languages (Python)